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QR Code-Based Ordering and Reservation System Design to Improve Cafe Operational Efficiency

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Abstract

The increasing demand for service efficiency and digital interaction in the food and beverage industry has driven businesses to adopt technological innovations that streamline ordering processes. Manual order-taking often results in long queues, inaccurate recording, and reduced customer satisfaction, emphasizing the need for digital transformation. This study aims to design and implement a QR Code-based ordering and reservation system that enhances operational efficiency and minimizes human error in café service management. The research employs a Prototyping development method consisting of five stages: data collection, system requirement analysis, interface design, prototype implementation, and functional testing. The system is developed as a web-based application that allows customers to place orders and make reservations independently through QR Code scanning. Testing results show that the system functions effectively across all core features, reducing customer waiting time and eliminating manual recording errors. The integrated reservation feature also improves table management and staff coordination. Unlike most prior QR Code-based systems that focus only on menu ordering, this study introduces an integrated, responsive web platform combining ordering and reservation in a unified workflow. The developed prototype addresses inefficiencies in manual service operations while demonstrating measurable improvements in speed, accuracy, and user satisfaction. This integration represents the main contribution of the research, providing a distinctive model for enhancing digital café operations through user-centered design. The study contributes to the advancement of digital ordering systems and offers both practical and academic implications for future F&B digitalization initiatives.

I. INTRODUCTION

The development of information technology in the digital era has significantly changed the way consumers interact with business services, including in the food and beverage (F&B) sector. The digitization of cafe services is now a strategic need, no longer an option, because customers expect a fast, efficient, and error-free process in every transaction. One form of such transformation is the implementation of a QR Code-based booking and reservation system, which allows customers to make orders independently using mobile devices. This innovation not only reduces queues and service delays but also improves the overall customer experience and operational consistency within cafe environments [1].

The object of this research focuses on Arka Coffee and Space Yogyakarta, a cafe that still uses a manual ordering and reservation system. The booking process is done directly through the waiter, while table reservations are made orally or through social media without systematic recording. This condition poses various problems, including long waiting times, the risk of recording errors, and the difficulty of recap daily order data. Therefore, a system is needed that can help customers make automatic bookings and reservations through a single, accessible web-based digital platform [2].

Implementing a QR Code-based digital system has proven to significantly increase operational efficiency and customer satisfaction across various F&B businesses. Empirical studies demonstrate that such systems can reduce average ordering time by 35% and increase transaction accuracy by up to 90%, emphasizing the role of digital solutions in enhancing both service speed and reliability [3]. In addition, digital ordering systems also allow for

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direct integration between customers, cashiers, and kitchens through a centralized data network, so that order information can be forwarded automatically and in real time [4]. This approach provides significant benefits to labor efficiency and raw material stock management, as well as creating a more modern and fast service experience for customers [5].

Nevertheless, although QR Code technology has been widely adopted, research gaps are still found in the context of integration between booking and reservation features in one single, web-based platform. Most previous research has focused only on digital menu ordering features without considering the table reservation aspect or has not been tested directly in the actual cafe operating environment [6], [7]. In addition, many systems developed do not prioritize responsive interface design, even though the user experience (UX) factor greatly determines the level of system adoption by customers [8]. This research is here to fill this gap by combining the two main functions of ordering and reservation into one QR Code-based web application that is implemented directly at Arka Coffee and Space Yogyakarta.

This research is important because the digitization of the ordering process has an impact not only on the speed of service, but also on the overall efficiency of cafe management. With the existence of a QR Code-based system, the management can monitor order activities, report recaps, and queue management in real time through the cashier dashboard. On the customer side, the use of this system increases independence in the ordering process without the need to wait for a server, while reducing the potential for miscommunication [9]. In addition, the implementation of this system supports the concept of contactless ordering services which are increasingly relevant after the COVID-19 pandemic, where customers tend to avoid direct interaction in the transaction process [10].

The main objective of this study is to design, implement, and evaluate a QR Code-based booking and reservation information system at Arka Coffee and Space Yogyakarta. To provide a sharper analytical direction, this research is guided by the following questions: (1) How can a web-based QR Code system be effectively designed to automate both ordering and reservation processes?, (2) To what extent can the system improve service efficiency, reduce human error, and enhance the user experience compared to manual workflows?, (3) How does the integration of ordering and reservation features contribute to the digital transformation of small-scale F&B businesses?. These questions provide the analytical foundation for developing, testing, and evaluating the system's practical and academic contributions. The results of this study are also expected to make a scientific contribution to the development of QR Code-based service information systems in the F&B sector and become a reference for small and medium enterprises in carrying out digital transformation [11].

Against this background, this research not only provides practical solutions for cafe management, but also expands academic insights on the application of QR Code technology in digital service management. The uniqueness of this research lies in its implementation in a fully operational cafe environment, the integration of ordering and reservation features in one responsive web system, and the application of a user-oriented Prototyping development model. Thus, this research has the potential to make an important contribution to the development of digital innovation in the field of cafe service management in Indonesia.

II. LITERATURE REVIEW

The digitalization of services in the food and beverage (F&B) sector has experienced rapid development in line with the increasing customer demand for fast, efficient, and minimal direct contact services. One of the innovations widely adopted in the restaurant industry is the QR Code-based ordering system, which allows customers to scan the code at the table to access digital menus, place orders, and send data directly to the cashier without manual interaction. The implementation of this system has been proven to speed up the ordering process, improve operational efficiency, and support a more hygienic and accurate customer experience in order recording [12].

However, despite the widespread use of QR Code-based ordering technology, most previous studies have primarily focused on digital menu ordering, while the integration of table reservation features and operational performance evaluation in real café environments has not been comprehensively addressed. This gap highlights the need for further investigation into how a unified web-based system can combine ordering and reservation functions to improve overall service efficiency. By identifying this research gap early, this study aims to establish a clearer direction for addressing both functional and managerial challenges in café operations.

Previous research has shown that the implementation of QR Codes in restaurant ordering systems can improve operational efficiency and customer satisfaction. Recent studies in the modern restaurant industry reveal that the use of QR menus is able to speed up the ordering process by reducing reliance on in-person staff interactions, thereby lowering customer wait times and improving the overall service experience [13]. Additionally, the use of QR Codes has also been shown to increase user satisfaction as it provides a fast and interactive service experience, as well as allowing customers to review orders before they are confirmed [14]. This strengthens the relevance of QR Code technology as a solution in the digital transformation of the culinary industry that is oriented towards customer satisfaction and service effectiveness.

From the technical side, several studies propose a web-based ordering system architecture that is integrated with QR Code as a connecting medium between customers and servers. These systems are generally developed using a combination of web technologies such as HTML, CSS, JavaScript, PHP, and MySQL to process orders in real time and forward them directly to the cashier to improve operational efficiency [15]. This kind of architecture proves to be efficient because it doesn't require any third-party add-ons or applications, making it easy to implement on different types of customer devices. However, most of the research still focuses on aspects of system functionality without addressing the integration of other features such as desk reservations or performance testing under high operational load conditions.

Empirical studies in Indonesia have also shown similar results. The implementation of a QR Code-based ordering system in local restaurants has been proven to be able to speed up the transaction process, improve the accuracy of order recording, and optimize the use of human resources in service operations [16]. Other research highlights aspects of the user interface (UI) and user experience (UX), where a simple and responsive system interface has been shown to increase user acceptance and reduce input errors during the booking process [17]. However, most studies are still limited to ordering systems without reservation features and have not conducted a thorough evaluation of system performance from a cafe management perspective.

Several recent studies highlight that the implementation of QR menus contributes to sustainable business practices while strengthening the hygienic perception of customers post-COVID-19 pandemic. QR Code-based digital systems are considered to be able to replace the use of physical menus, reduce the potential for contamination, and increase customer safety and comfort when interacting in the service area [18], [19]. In addition, this approach also supports the environmentally friendly aspect by reducing paper use and operational costs related to menu printing. However, there have not been many studies that directly link these benefits to measuring operational efficiency and the rate of order recording errors quantitatively. Therefore, this study seeks to address the identified gaps by designing and implementing a fully integrated QR Code-based ordering and reservation system tested in a real café setting. This approach not only contributes to improving service efficiency and accuracy but also provides empirical validation of the system's performance in operational contexts where both human and technical factors interact.

From the various literature above, it can be identified that research gaps are still open. First, most studies only highlight QR Code-based ordering systems without integration with table reservation features that are important in the context of cafe operations. Second, previous research has focused more on the technical implementation and appearance of the interface, while measures of the effectiveness of the system on cafe operational performance (such as service time and recording error rates) are still rare. Third, there have not been many studies that have carried out direct implementation in a fully operational cafe environment to assess the performance of the system under actual conditions. Therefore, this study is here to fill this gap by designing and implementing an integrated QR Code-based booking and reservation system, as well as evaluating its impact on service efficiency and the accuracy of booking data in a real environment. Thus, the results of this study are expected to make a dual contribution, namely practically helping to improve the effectiveness of services in cafes, and academically enriching the literature related to the implementation of QR Code-based information systems in the context of modern culinary business. The findings of this research are also expected to be a reference for system developers in designing web-based applications that combine integrated booking, reservation, and transaction management features.

III. METHODS

This research uses the Prototyping method as a system development approach because it has an iterative and flexible nature, which allows developers to make improvements based on direct feedback from users. This model is ideal for the development of responsive web applications that require quick validation of interface functions and displays. The Prototyping approach was chosen because it can accelerate the system development process to the trial stage, as well as allow direct testing of the User Experience (UX) and User Interface (UI) aspects of users, both customers and cafe management [1]. The main technology application in this study includes HTML, CSS, and JavaScript for the frontend side, as well as PHP and MySQL for the backend side, to ensure that the system is easy to implement, efficient, and in accordance with the resource capacity at Arka Coffee and Space Yogyakarta [2]. The stages of the research are arranged chronologically and systematically following the sequence of steps in the Prototyping model as shown in Figure 1.

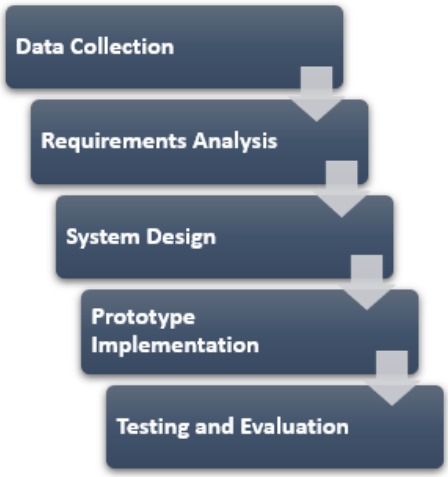


Fig 1. Research Flow Using Prototyping Method

Figure 1 illustrates the sequence of research steps starting from data collection to the system evaluation stage. Each stage is interconnected sequentially, where the results of one stage become the basis for the implementation of the next stage. This series of processes ensures that the developed system truly represents the needs of users and can function optimally in the operational context of Arka Coffee and Space Yogyakarta. The following are the details of the stages of the research conducted:

A. Data Collection

The initial stage of the research was carried out by collecting primary data through direct observation and semi-structured interviews with the management of Arka Coffee and Space during the period from March to July 2025. The observation aims to study in detail the flow of manual ordering and reservations implemented in the cafe, including customer waiting times, the process of interaction with waiters, and manual order recording. From the observations, it was found that the manual process often causes long queues, delays in delivering orders to the kitchen, and recording errors due to human error. The interviews were conducted to identify more efficient system needs, such as self-ordering by customers via QR Codes, integration of order notifications to cashier and kitchen, and system-based automated desk reservation features.

B. System Needs Analysis

After the data is obtained, a system needs analysis is carried out to break down the results of observations and interviews into functional and non-functional specifications. This analysis helps determine the data structure, processing flow, and key features that must be available in the system in order to solve the operational constraints faced by the cafe. The functional needs identified include customer login features, QR Code scanning, digital menu ordering, review orders, and automatic recording to the cashier dashboard. The non-functional needs include interface responsiveness, ease of use on mobile devices, and system access speed during operating hours. With this comprehensive needs analysis, the system design can be focused on improving service efficiency and order information accuracy [22].

C. System Design

The system design stage is carried out using conceptual and structural modeling to describe the user's interaction with the system. Conceptual modeling is presented in the form of a flowchart, which describes the sequence of the process from the scanning of the QR Code by the customer to the delivery of the order data to the cashier dashboard. Meanwhile, the Use Case Diagram is used to show the relationships between customer actors, cashiers, and systems in each of the application's key functions. This design is done to ensure that the system has an efficient workflow, is easy for users to understand, and is the main guideline in the implementation process. This modeling approach has been proven to improve design consistency, reduce errors at the development stage, and ensure the fit between the system design and the actual needs of the user [4].

D. Prototype Implementation

The implementation stage begins with the development of a prototype of a responsive web application that can be accessed through a mobile browser without the need for additional installation. The system is developed using HTML, CSS, and JavaScript for the frontend side, as well as PHP and MySQL for the backend side, which

function to organize business logic and store order data temporarily (real-time session). The main focus of the implementation is on two crucial functions, namely QR Code-based digital ordering and order management on the cashier dashboard. Through this system, customers can simply scan the QR Code available at each table to display the menu, select food and drinks, add special notes, and review the order before confirming. Order data is then automatically forwarded to the cashier dashboard, making it easier to process orders and reduce manual recording errors.

E. Prototype Testing and Evaluation

The final stage of the research is the testing and evaluation of the prototype using the Black Box Testing method, which focuses on the compatibility between the input and output without paying attention to the internal structure of the code [5]. This test covers several important aspects such as the validity of the QR Code scan on each table, the accuracy of the calculation of the total cost of the order, and the success of the data transfer to the cashier dashboard. All key features are tested to ensure they work according to the user's needs. The test results show that the system is able to carry out the login, order, review, and data delivery processes well, according to the specifications that have been set. This stage also includes an overall evaluation of the prototype's performance to assess the reliability and ease of use of the system, as well as being the basis for the development of additional features such as digital payment integration at the next stage of research [25].

Through this series of stages, this research succeeded in producing a prototype of a QR Code-based booking and reservation application that functions effectively and in accordance with the operational needs of Arka Coffee and Space Yogyakarta. This system is able to improve service efficiency by speeding up the ordering process, reducing customer queues, and reducing the rate of recording errors that previously often occurred in manual methods. In addition, the implementation of QR Code technology allows customers to place orders independently without direct interaction with the server, thus speeding up the flow of services and providing a more practical and modern experience. These results show that the implementation of web-based digitization and QR Codes not only simplifies the operational process for management, but also increases overall customer comfort and satisfaction.

IV. RESULTS

The results of this study are in the form of a prototype of a QR Code-based booking and reservation information system that was implemented and tested in the Arka Coffee and Space Yogyakarta environment. The system was developed using web technology with the aim of improving operational efficiency, reducing customer queues, and minimizing order recording errors that often occur in manual methods. This section describes the results of the research in three main aspects, namely the implementation of the system prototype, the results of functional testing, and the evaluation of the system implementation.

A. Prototype Implementation

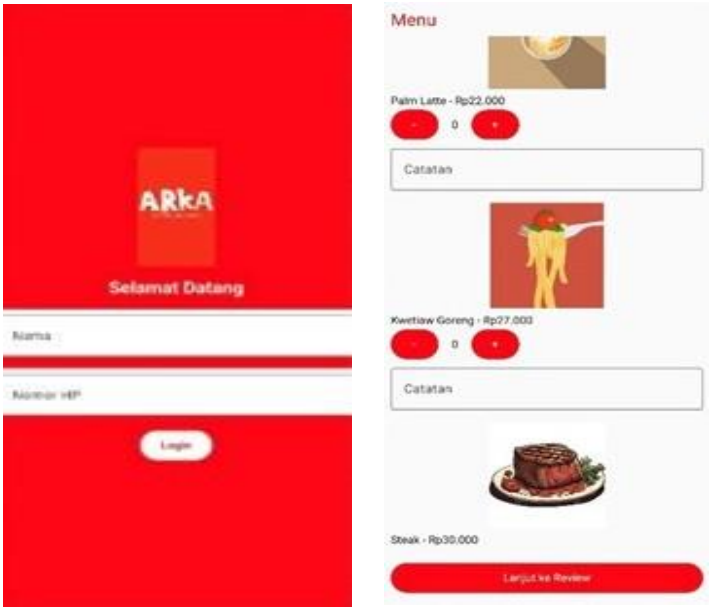
The system prototype was developed using HTML, CSS, and JavaScript on the frontend side and PHP and MySQL on the backend side. The system is designed to be accessible through a mobile browser without the need for additional application installation, so customers can simply scan the QR Code available at the table to start the ordering process. The system interface is made responsive and intuitive so that it can be easily used by customers from various devices. The stages of use start with the initial view (splash screen) and the login page. As shown in Figure 2(a), the home page displays the Arka Coffee and Space logo as the app's visual identity. Furthermore, in Figure 2(b), users are asked to enter their name and mobile phone number as temporary data during the booking session. This process serves to initialize the user's session and make it easy to track the transaction history in the system.



(a) (b)

Fig. 2 (a) Application Splash Screen and (b) User Login Page

After successfully logging in, the user is redirected to the main page with the "Scan QR" button, as shown in Figure 3(a). This feature is used to scan the QR Code located on each customer's desk. The scan results connect the user directly with the corresponding table data in the system database. Once the QR Code has been successfully scanned, the system displays a digital menu page, as seen in Figure 3(b). This page contains a complete list of foods and beverages with prices, images, and additional note fields, so customers can add items as needed and adjust orders before they are sent to the system.



(a) (b)

Fig. 3 (a) QR Code Scan Button and (b) Digital Menu Page

Once the customer has finished selecting the menu, the system displays the Order Review page to ensure that each item selected has been matched to the user's wishes. The review page view is shown in Figure 4, which shows a complete list of orders with quantities, additional notes, and total provisional prices. On this page, users can review all order items, delete or add menus, and update notes before pressing the confirmation button. This stage serves as a final verification to minimize order errors before data is sent to the main system.

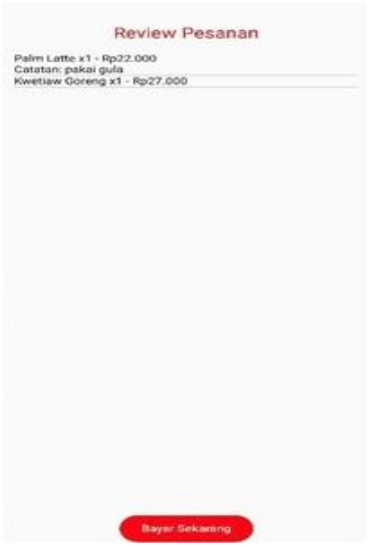
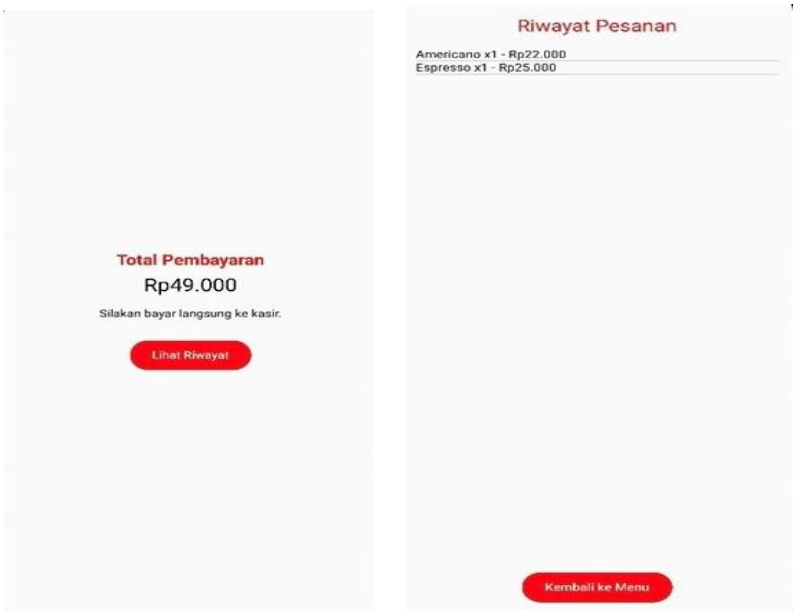


Fig. 4 Order Review Page View

Once the user presses the confirmation button on the review page, the system automatically displays the total payment page as seen in Figure 5(a). This page displays the total cost based on all confirmed order items. Although the system is not yet integrated with the digital payment feature, the total price displayed serves as a reference for manual payments at the cashier. In addition, the system also provides a "View History" feature as shown in Figure 5(b), which displays a list of orders that have been placed during an active session. This feature makes it easier for customers to review orders and assist cashier staff in verifying ongoing transactions.



(a) (b)

Fig. 5 (a) Total Payment Page and (b) Order History Page

If users want to make additional orders or just browse the menu again, they can press the "Return to Menu" button. This button will redirect the user to the start menu page as shown in Figure 3(b). without deleting the history of previous orders while the session is still ongoing. These features play an important role in maintaining data transparency and accuracy between customers and the cashier. All order data sent through the system will be automatically forwarded to the cashier dashboard without the need for manual recording by the server. Thus, the ordering process becomes faster, more structured, and reduces the potential for order recording errors.

B. Prototype Test Results

To ensure that the system runs as needed, functional testing is carried out using the Black Box Testing method. Testing is done by independently testing each key feature to verify that the system output matches user feedback. The aspects tested included customer login, QR Code scanning, menu ordering, order review process, payment

confirmation, and data submission to the cashier dashboard. The test results are summarized in Table 1, which shows that all the main features of the system have been functioning properly. Each test produces an output that conforms to the specifications, indicating that the system is working stably and responsive to a wide range of user inputs.

TABLE 1
FUNCTIONAL TEST RESULTS OF OR CODE-BASED ORDERING SYSTEM APPLICATION

Feature	Tested Inputs	Expected Output	Test Results
Login Page	Input of valid name and mobile number	The user successfully logged in and received a personal greeting	**Succeed**
Login Page	Empty or invalid inputs	Error notification appears and cannot continue	**Succeed**
QR Code Scanning	Scanning the QR code on the table	The system displays menu pages according to the scanned table	**Succeed**
Menu Page	Add an item to an order list	Items appear in the order list with quantity and record	**Succeed**
Menu Page	Remove an item from the order list	Items removed from the order list	**Succeed**
Order Review Page	Pressing the "Continue to Preview" button	View entire order complete with quantity and record	**Succeed**

Based on Table 1, the test results show that the system is able to handle all key functions consistently without interruption. The QR Code scanning process can be done quickly, the calculation of the total order price is accurate, and the order data is successfully sent to the cashier dashboard automatically. In addition, no significant system errors (error handling) were found during the testing process. This proves that the developed prototype has met all expected functional criteria.

C. System Implementation Evaluation

The evaluation was carried out to assess the effectiveness and ease of use of the system both from the user and management sides. Based on the results of observations during implementation, the QR Code-based ordering and reservation system has been proven to be able to speed up the ordering process and significantly reduce customer queues compared to manual methods. Through the digital ordering feature, customers can interact directly with the menu without the need to wait for a waiter, thereby improving service time efficiency and user convenience. The system's simple, responsive, and easy-to-understand interface also makes the app usable by customers of all ages without the need for special training.

From an operational perspective, the implementation of this system helps management in reducing recording errors and accelerating the recapitulation process of daily order reports. Every transaction that enters the cashier dashboard is automatically saved and can be used for operational analysis, thus supporting data-driven decision-making. Overall, the implementation of this digital system is able to improve service efficiency, improve recording accuracy, and provide a faster, practical, and modern ordering experience for customers. Digitizing web-based services like this can be a strategic solution for culinary business actors to improve service quality and competitiveness in the information technology era.

V. DISCUSSION

The results of the system implementation show that the implementation of a QR Code-based booking and reservation system significantly improves the efficiency of the service process and reduces order recording errors compared to the manual method. Customers can place orders and reservations independently without the need to wait for a waiter, reducing waiting time and queues in the cashier area. This condition is in line with research that states that the implementation of QR Code-based digital menus is able to speed up the transaction process and provide a more efficient service experience for customers in the F&B sector [26]. In addition, the system developed has a high level of responsiveness and can be accessed optimally through mobile devices, thus supporting the concept of contactless service which is a major trend in improving user experience post-COVID-19 pandemic [27].

From the results of the functional test, all the key features from user login, QR Code scanning, menu ordering, to order confirmation went well without any system errors found. The data communication process between application modules takes place in real time, ensuring that customer orders are directly forwarded to the cashier dashboard. This supports the findings of similar studies that the success of digital ordering systems relies heavily on reliable data communication and intuitive user interface design [28]. Additionally, the order review feature that allows customers to review orders before they are confirmed has been proven to improve data accuracy and user satisfaction.

Analysis from the operational side shows that the implementation of this system not only increases the speed of service, but also reduces the workload of employees, especially in the process of recording and managing

orders. The use of QR Codes also has a positive impact on business sustainability by reducing paper use and minimizing the risk of errors due to manual interactions. This is in line with the results of the study which highlights that the digitization of QR Code-based services provides long-term benefits to operational efficiency and environmental sustainability because it reduces operational costs and increases customer satisfaction [29]. Thus, the system developed in this study is able to provide practical solutions that are adaptive to the needs of the modern culinary industry.

However, this study has several limitations that need to be considered. First, the tests were conducted at only one cafe location, so efficiency results could not be generalized for different types of F&B businesses with different capacities and operational patterns. This single-location testing may also introduce potential bias, as the demographic characteristics of Arka Coffee and Space Yogyakarta customers such as age group, digital literacy level, and visit patterns may not represent the wider customer base of other cafes. In addition, the unique operational conditions of the cafe, including staff management and service layout, could influence the perceived performance of the system, potentially affecting the external validity of the results. Second, the system has not been integrated with digital payment modules (e-payments), which can actually increase customer convenience and transaction efficiency. Third, this study does not include simultaneous testing of system performance with a high number of users, so the system's ability to handle peak loads is still empirically unknown. Future research is expected to expand testing on a larger scale by adding load testing analysis and API-based payment method integration.

In addition, further research is suggested to examine user behavior factors and customer loyalty towards QR Code-based digital systems. Recent studies show that customer perceptions of the convenience and usefulness of using QR Codes have an effect on adoption and loyalty rates, with results varying between user groups [30]. This shows the importance of understanding aspects of user behavior, such as trust, perceived usefulness, and effort expectancy, in determining the success of similar system implementations in the future. By adding the Technology Acceptance Model (TAM) or Unified Theory of Acceptance and Use of Technology (UTAUT) model approach to subsequent research, developers can gain a deeper understanding of the level of customer acceptance of digital service technology. Overall, the results of this study strengthen the literature that confirms that the integration of QR Code-based booking and reservation systems can be an effective digital solution to improve service efficiency, customer satisfaction, and business sustainability in the F&B sector.

VI. CONCLUSIONS

This research successfully designed and implemented an integrated QR Code-based ordering and reservation system to improve service efficiency and order recording accuracy in cafe operations. This system is able to answer the main problems identified in the study, namely the length of customer waiting time, the high risk of manual recording errors, and the limitations of the conventional reservation system. With a Prototyping development approach, the system is tested through several stages and shows results that suit the user's needs. The results of the functional tests show that all key features, including QR Code scanning, menu ordering, order confirmation, and transaction history storage, can run well and provide a responsive and efficient user experience. Thus, this study provides empirical evidence that digitizing the ordering process through web-based QR Code integration can be an effective solution to increase service speed and reduce employee workload in the culinary business. However, the scalability of the system still presents potential challenges, such as variations in device compatibility, network stability, and possible server load during peak usage periods, which need to be carefully addressed in future deployments. Recognizing these limitations ensures a balanced view of the system’s current performance and its readiness for broader adoption.

The main contribution of this research lies in the development of an integrated system model that combines booking and reservation features in a single web-based digital platform, which has not been extensively discussed in the previous literature. In addition, this study expands the understanding of the application of QR Code technology in the context of real cafe operations, not just simulations or laboratory tests. This research also provides an empirical basis for further studies to develop similar systems with a wider scope, such as digital payment integration, load testing for performance testing under high load conditions, and user behavior analysis using the TAM or UTAUT approach. Future research is also encouraged to explore cross-platform development (e.g., mobile-native and progressive web applications) to enhance accessibility and performance across diverse devices. In addition, longitudinal evaluation can be conducted to assess long-term user adoption, behavioral consistency, and the sustainability of system utilization in real operational environments. Overall, this study answers the main question that the implementation of a QR Code-based ordering and reservation system can improve operational efficiency, improve recording accuracy, and make a real contribution to digital transformation in the modern food and beverage sector.

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